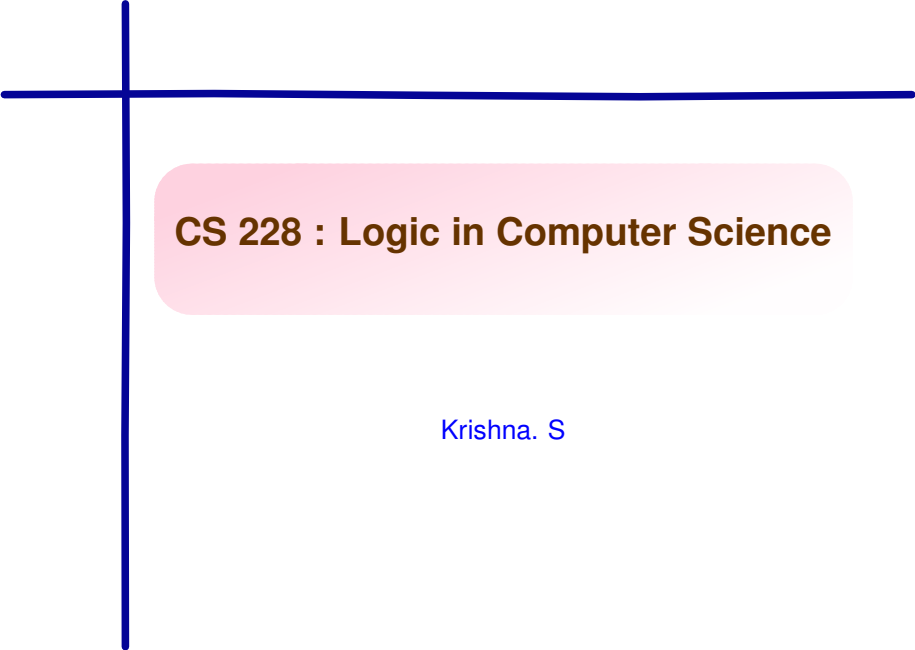




CS 228 : Logic in Computer Science

Krishna. S

So Far

- ▶ ω -automata with Büchi acceptance, also called Büchi automata
- ▶ Non-determinism versus determinism

Büchi Acceptance

A language $L \subseteq \Sigma^\omega$ is called ω -regular if there exists a NBA $\mathcal{A}$ such that $L = L(\mathcal{A})$.

Emptiness

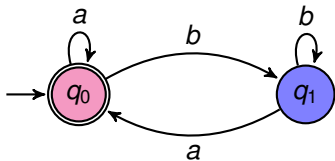
Given an NBA/DBA $\mathcal{A}$, how do you check if $L(\mathcal{A}) = \emptyset$?

Emptiness

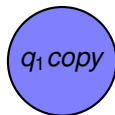
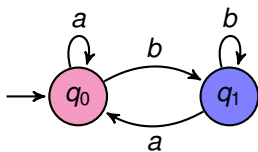
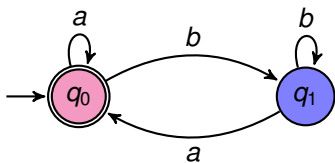
Given an NBA/DBA $\mathcal{A}$, how do you check if $L(\mathcal{A}) = \emptyset$?

- ▶ Enumerate SCCs
- ▶ Check if there is an SCC containing a good state

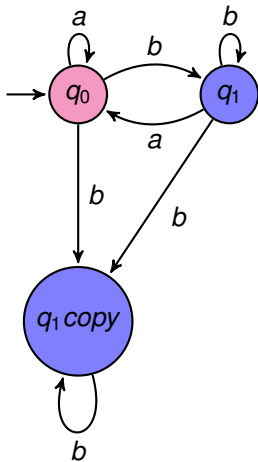
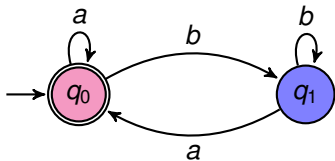
Complementation of DBA



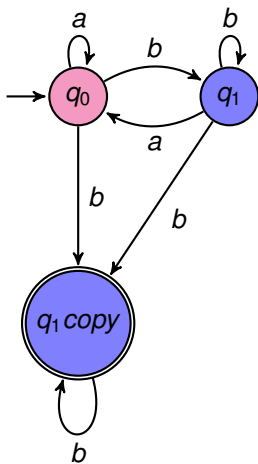
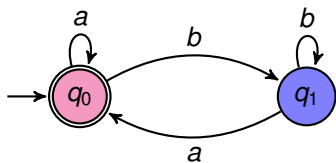
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Complementation of DBA



Complementation of DBA

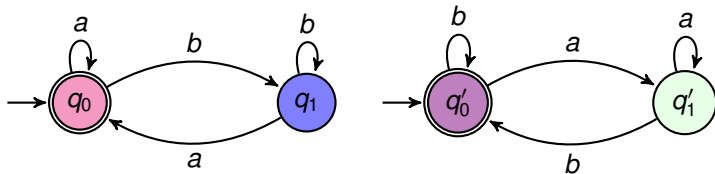


Complementation of DBA

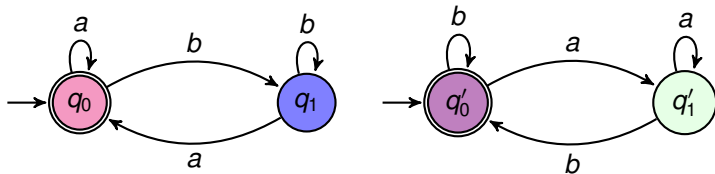
- ▶ Given $\mathcal{A}$ is a DBA, and $w \notin L(\mathcal{A})$, then after some finite prefix, the unique run of w settles in bad states.
- ▶ Idea for complement: “copy” states of $Q - G$, once you enter this block, you stay there.
- ▶ View this as the set of good states, any word w that was rejected by $\mathcal{A}$ has two possible runs in this automaton: the original run, and one another, that will settle in the $Q - G$ copy, and will be accepted.
- ▶ What we get now is an NBA for $\overline{L(\mathcal{A})}$, not a DBA.

Complementing NBA non-trivial, can be done.

Union and Intersection of NBA

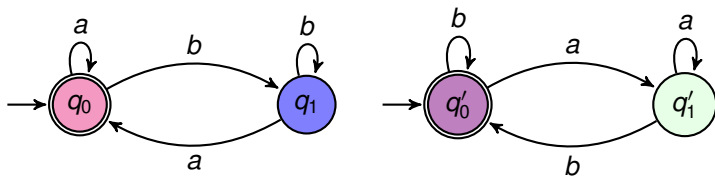


Union and Intersection of NBA



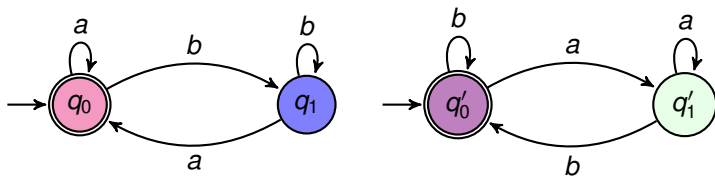
- States as $Q_1 \times Q_2 \times \{1, 2\}$, start state $(q_0, q'_0, 1)$

Union and Intersection of NBA



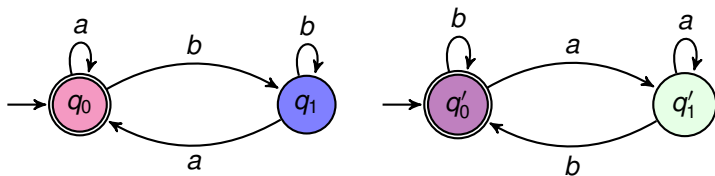
- ▶ States as $Q_1 \times Q_2 \times \{1, 2\}$, start state $(q_0, q'_0, 1)$
- ▶ $(q_1, q_2, 1) \xrightarrow{a} (q'_1, q'_2, 1)$ if $q_1 \xrightarrow{a} q'_1$ and $q_2 \xrightarrow{a} q'_2$ and $q_1 \notin G_1$
- ▶ $(q_1, q_2, 1) \xrightarrow{a} (q'_1, q'_2, 2)$ if $q_1 \xrightarrow{a} q'_1$ and $q_2 \xrightarrow{a} q'_2$ and $q_1 \in G_1$

Union and Intersection of NBA



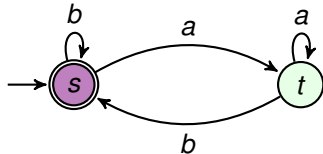
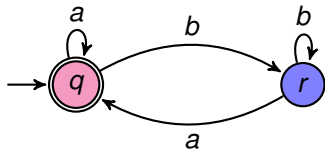
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Union and Intersection of NBA

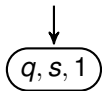
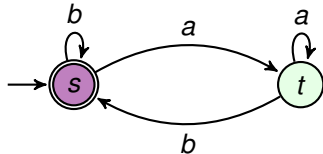
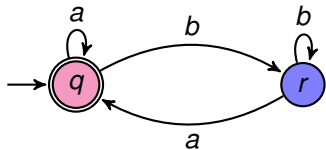


- ▶ States as $Q_1 \times Q_2 \times \{1, 2\}$, start state $(q_0, q'_0, 1)$
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- ▶ Good states = $Q_1 \times G_2 \times \{2\}$ or $G_1 \times Q_2 \times \{1\}$

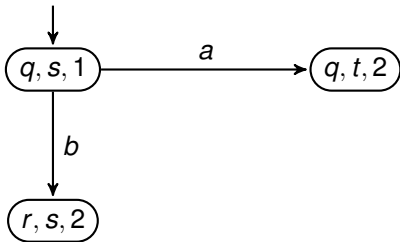
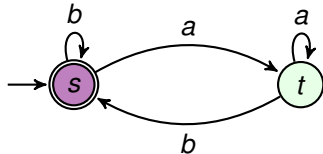
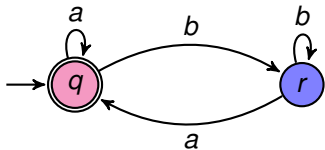
Union and Intersection of NBA



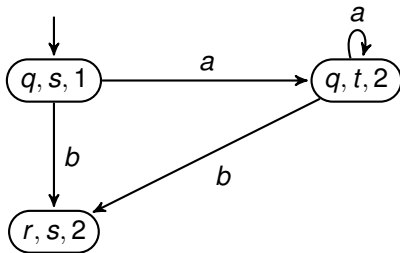
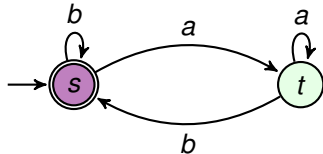
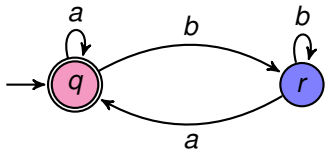
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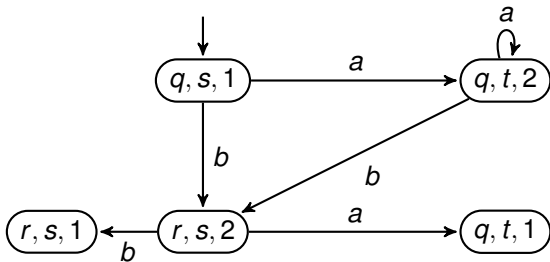
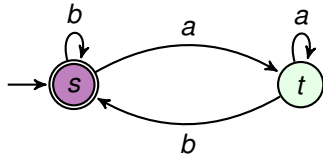
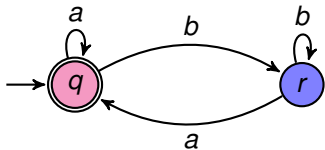
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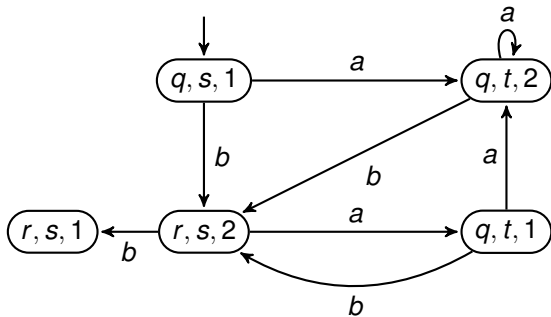
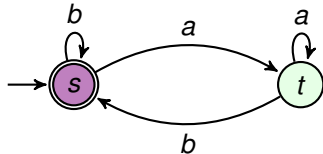
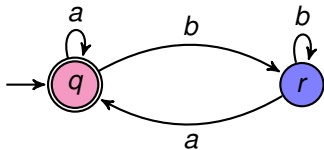
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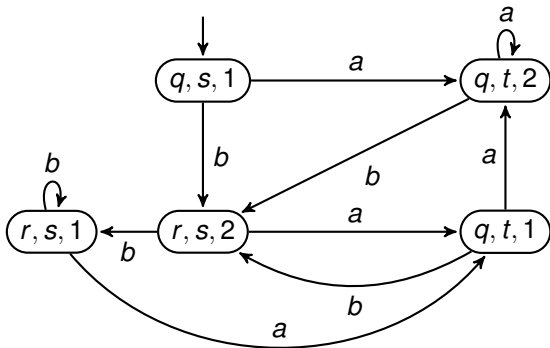
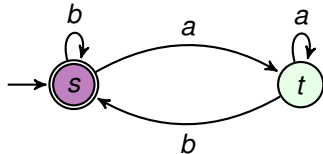
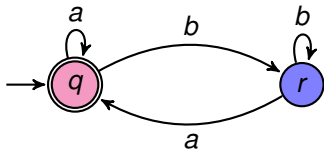
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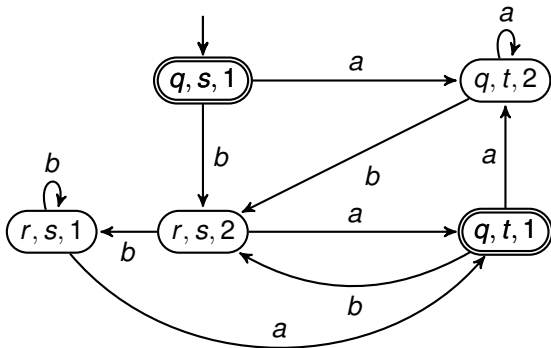
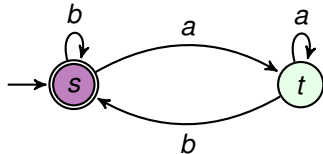
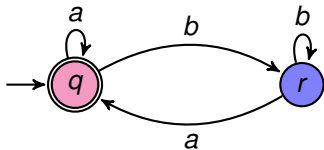
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Union and Intersection of NBA



Union and Intersection of NBA



Normal Form for ω -regular languages

An ω -regular language $L \subseteq \Sigma^\omega$ can be written as $L = \bigcup_{i=1}^n U_i V_i^\omega$, where U_i, V_i are regular languages.

One direction : Assume L is accepted by an NBA/DBA.

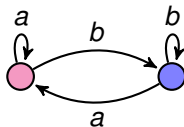
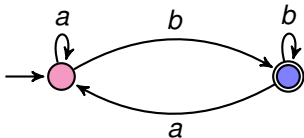
- ▶ Define $U_g = \{w \in \Sigma^* \mid q_0 \xrightarrow{w} g\}$
- ▶ Define $V_g = \{w \in \Sigma^* \mid g \xrightarrow{w} g\}$
- ▶ Then $L = \bigcup_{g \in G} U_g V_g^\omega$, where U_g, V_g are regular
- ▶ Show that U_g, V_g are regular.

Normal Form for ω -regular languages

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Other direction : Assume $L = \bigcup_{i=1}^n U_i V_i^\omega$. Show that L is accepted by an NBA/DBA.

1. If V is regular, V^ω is ω -regular

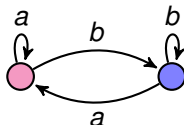
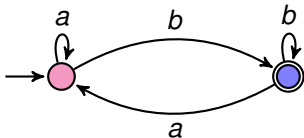


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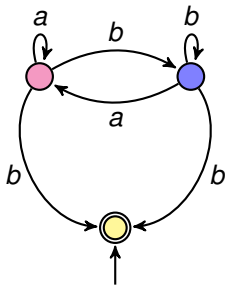
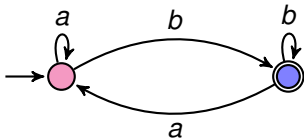


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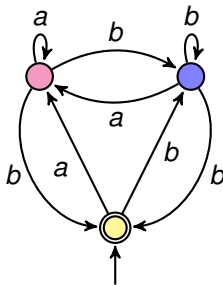
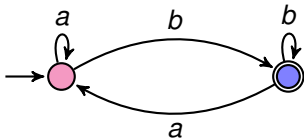


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Normal Form for ω -regular languages

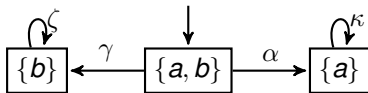
1. If V is regular, V^ω is ω -regular
 - ▶ Let $D = (Q, \Sigma, q_0, \delta, F)$ be a DFA accepting V
 - ▶ Construct NBA $E = (Q \cup \{p_0\}, \Sigma, p_0, \Delta, G)$ such that $G = \{p_0\}$,
 - ▶ $\Delta = \delta \cup \{p_0 \in \Delta(q, a) \mid \delta(q, a) \in F\} \cup \{\Delta(p_0, a) = s \mid \delta(q_0, a) = s\}$
2. Show that if U is regular and V^ω is ω -regular, then UV^ω is ω -regular
 - ▶ $D = (Q_1, \Sigma, q_0, \delta_1, F)$ be a DFA, $L(D) = U$ and $E = (Q_2, \Sigma, q'_0, \delta_2, G)$ be an NBA, $L(E) = V^\omega$.
 - ▶ $A = (Q_1 \cup Q_2, \Sigma, q_0, \delta', G)$ NBA such that $\delta' = \delta_1 \cup \delta_2 \cup \{(q, a, q'_0) \mid \delta_1(q, a) \in F\}$

LTL ModelChecking

- ▶ Given transition system TS , and LTL formula φ , does $TS \models \varphi$?
- ▶ $Tr(TS) \subseteq L(\varphi)$ iff $Tr(TS) \cap \overline{L(\varphi)} = \emptyset$
- ▶ First construct NBA $A_{\neg\varphi}$ for $\neg\varphi$.
- ▶ Construct product of TS and $A_{\neg\varphi}$, obtaining a new TS, say TS' .
- ▶ Check some very simple property on TS' , to check $TS \models \varphi$.

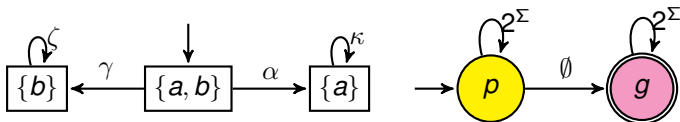
An Example $TS \models \varphi$

- ▶ Let $\varphi = \Box(a \vee b)$, $\neg\varphi = \Diamond(\neg a \wedge \neg b)$
- ▶ Take TS and $A_{\neg\varphi}$, and check the intersection.



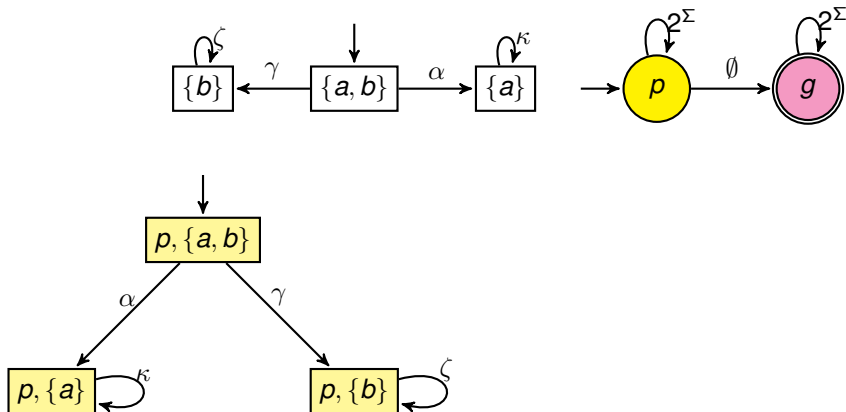
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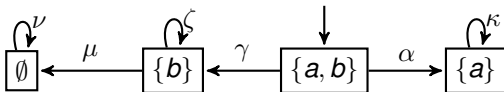
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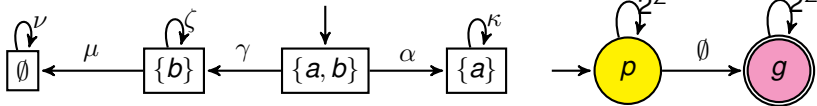
An Example : $TS \not\models \varphi$

- ▶ Let $\varphi = \Box(a \vee b)$, $\neg\varphi = \Diamond(\neg a \wedge \neg b)$
- ▶ Take TS and $A_{\neg\varphi}$, and check the intersection.
- ▶ **Alert : Missing transition in TS'**



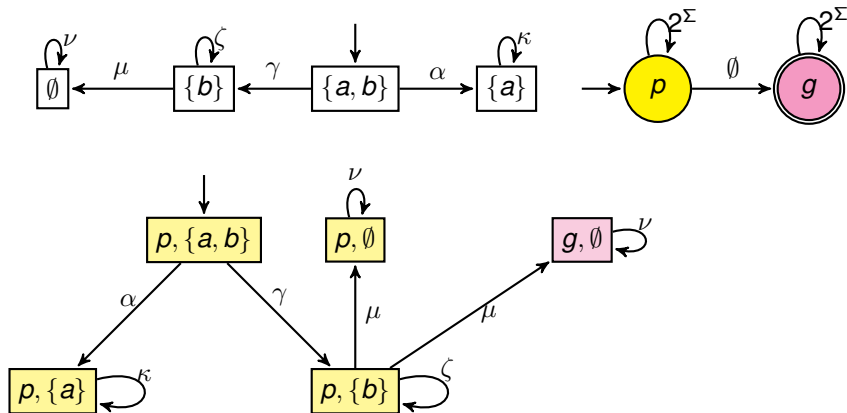
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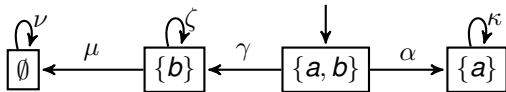
Product of TS and NBA

Given $TS = (S, Act, I, AP, L)$ and $\mathcal{A} = (Q, 2^{AP}, \delta, Q_0, G)$ NBA.
Define $TS \otimes \mathcal{A} = (S \times Q, Act, I', AP', L')$ such that

- ▶ $I' = \{(s_0, q) \mid s_0 \in I \text{ and } \exists q_0 \in Q_0, q_0 \xrightarrow{L(s_0)} q\}$
- ▶ $AP' = Q, L' : S \times Q \rightarrow 2^Q$ such that $L'((s, q)) = \{q\}$
- ▶ If $s \xrightarrow{\alpha} t$ and $q \xrightarrow{L(t)} p$, then $(s, q) \xrightarrow{\alpha} (t, p)$

Persistence Properties

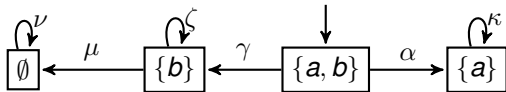
Let η be a propositional logic formula over AP . A persistence property P_{pers} has the form $\Diamond\Box\eta$. How will you check a persistence property on a TS?



- ▶ For example, $TS \not\models \Diamond\Box(a \vee b)$
- ▶ For example, $TS \models \Diamond\Box(a \vee (a \rightarrow b))$

Persistence Properties

Let η be a propositional logic formula over AP . A persistence property P_{pers} has the form $\Diamond\Box\eta$. How will you check a persistence property on a TS?



- ▶ For example, $TS \not\models \Diamond\Box(a \vee b)$
- ▶ For example, $TS \models \Diamond\Box(a \vee (a \rightarrow b))$
- ▶ $TS \not\models P_{pers}$ iff there is a reachable cycle in the TS containing a state with a label which satisfies $\neg\eta$.

LTL ModelChecking

- ▶ Given TS and LTL formula φ . Does $TS \models \varphi$?
- ▶ Construct $A_{\neg\varphi}$, and let $g_1, \dots, g_n$ be the good states in $A_{\neg\varphi}$.
- ▶ Build $TS' = TS \otimes A_{\neg\varphi}$.
- ▶ The labels of TS' are the state names of $A_{\neg\varphi}$.
- ▶ Check if $TS' \models \Diamond\Box(\neg g_1 \wedge \dots \neg g_n)$. (check the example!)

LTL ModelChecking

- ▶ Given TS and LTL formula φ . Does $TS \models \varphi$?
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ModelChecking LTL in TS = Check Persistence in TS'

The following are equivalent.

- ▶ $TS \models \varphi$
- ▶ $Tr(TS) \cap L(A_{\neg\varphi}) = \emptyset$
- ▶ $TS' \models \Diamond\Box(\neg g_1 \wedge \dots \neg g_n)$.

Complexity of LTL Modelchecking

TBD after LTL satisfiability